

Support Vector Machines (SVM): An Overview

Support Vector Machines (SVM) are powerful algorithms used to classify data by finding the optimal line or boundary that separates different groups. This report explores how SVMs work, their relationship with cross-validation, and the mathematical concepts behind them.

Cross-Validation: Evaluating Model Performance

Cross-validation is a technique that allows us to compare different machine learning models and assess how well they are likely to perform in practice. It involves dividing the dataset into different parts (or *folds*) and testing the model's performance on each fold. This ensures that the model is evaluated on multiple subsets of data, giving a clearer sense of its accuracy.

- **Four-Fold Cross-Validation:** The dataset is split into 4 blocks. Each block is tested individually, while the other blocks are used for training.
- **Leave-One-Out Cross-Validation:** Each data point is tested individually, leaving out one observation at a time.
- **Confusion Matrix:** For each method, a confusion matrix is used to analyse the model's performance. It tells us what the algorithm predicted correctly and where it went wrong.

Threshold, Margin, and Maximal Margin Classifier (MMC)

In SVM, a **threshold** is defined to classify data into groups. The threshold can often be a midpoint that separates the two classes. The shortest distance between any observation and the threshold is called the **margin**. A larger margin implies a better-defined separation between classes.

When the threshold is chosen to maximize the margin, the algorithm becomes a **Maximal Margin Classifier (MMC)**. However, MMC has a major drawback:

- It is extremely sensitive to *outliers* in the training data.

Bias-Variance Trade-off

The **bias-variance trade-off** explains the balance between two types of errors:

- **Bias:** Error due to over approximation of the problem .High bias occurs when the model makes strong assumptions about the data(underfitting).
- **Variance:** Error caused by sensitivity to small fluctuations in the training data (overfitting). High variance occurs when the model learns the noise in the data too well.

Relationship:

- As model complexity increases, bias decreases, but variance increases.
- A good model finds a balance between bias and variance.

To address the outlier problem and improve generalization and obtain a threshold that is not so sensitive to the outliers, we allow some **misclassifications**. This approach is called **Soft Margin Classification**.

Soft Margin Classifier and Support Vectors

When misclassifications are allowed, the threshold can adapt better to the data. In this case:

- The margin between observations and the threshold becomes a **soft margin**.
- The classifier that uses this margin is called a **Support Vector Classifier (SVC)**.
- Observations on the edge of or within the soft margin are called **support vectors**.

SVC Across Dimensions:

- In 1D: The SVC is a single point on the number line.
- In 2D: The SVC is a line.
- In 3D: The SVC is a plane.
- In higher dimensions (4D and above): The SVC is a **hyperplane**.

A hyperplane is essentially a flat affine subspace that separates data in higher dimensions.

Limitations of SVC and the Introduction of SVM

While SVC allows misclassifications, it struggles when the data has many overlaps, leading to frequent misclassifications. To solve this issue, we use **Support Vector Machines (SVM)**.

How SVM Works:

1. Start with data in a relatively low-dimensional space.
2. Move the data into a higher-dimensional space.
3. Find the **optimal hyperplane** (SVC) that separates the data in the higher dimension.

Kernel Functions and the Kernel Trick

The question arises: how do we move the data into higher dimensions efficiently? This is where **kernel functions** come into play.

- A kernel function computes the relationship between each pair of observations as if they were in a higher dimension, without explicitly transforming the data.
- This method is called the **kernel trick**. It allows SVM to avoid the computational burden of transforming the data into higher dimensions while still achieving the same results.
- **Key Advantage:** The kernel trick reduces computational steps and enables calculations in *infinite dimensions* using functions like the radial kernel.

Types of Kernel Functions

1. Polynomial Kernel

The polynomial kernel transforms the data into higher dimensions using the following equation:

general form: $(\mathbf{a} \times \mathbf{b} + \mathbf{r})^d$

- $\mathbf{a}$ and $\mathbf{b}$ are points in the dataset.
- $\mathbf{r}$ is a coefficient.
- $\mathbf{d}$ is the degree of the polynomial.
- Both $\mathbf{r}$ and $\mathbf{d}$ are determined using cross-validation.

For example, let's assume $\mathbf{d} = 2$ and $\mathbf{r} = 1/2$:

given: $(\mathbf{a} \times \mathbf{b} + 1/2)^2$

Expanding:

$$= (\mathbf{a} \times \mathbf{b} + 1/2)(\mathbf{a} \times \mathbf{b} + 1/2)$$

$$= (\mathbf{a} \times \mathbf{b})^2 + (\mathbf{a} \times \mathbf{b}) + 1/4$$

This equation represents the coordinates of the transformed data in higher dimensions (ignoring redundant axes).

2. Radial Basis Function (RBF) - Radial Kernel

The **RBF kernel** computes the SVC in infinite dimensions. It behaves like a weighted **nearest-neighbour model**:

- Nearby observations have a stronger influence on the classification of a new point.
- The radial kernel equation is:

given: $e^{-(\mathbf{a} - \mathbf{b})^2}$

Here:

- $a - b$ represents the distance between two points.
- γ determines how much influence neighbouring points have. It is optimized using cross-validation.

Key Insight:

- If two points are far apart, the value of the radial kernel approaches zero, meaning they have little influence on each other.

NOTE: The **RBF kernel** computes the SVC in infinite dimensions. This can be showed by applying Taylor series on e^{ab} and simplifying further.

Conclusion

Support Vector Machines (SVM) are powerful tools for classification, capable of handling complex, high-dimensional datasets. By using techniques like soft margins, support vectors, and kernels (like polynomial or radial), SVM finds optimal decision boundaries. Cross-validation helps fine-tune the model, ensuring it performs well on unseen data by balancing bias and variance.